Supplementary Material

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1 Supplementary Data

1.1 List 1: Search strategy for MEDLINE and EMBASE

1 Activated Protein C Resistance/

2 exp Factor V/ or blood clotting factor 5/ or blood clotting factor 5 leiden/

3 Thrombophilia/co, cn, ge, pp, bl

4 Polymorphism, Single Nucleotide/ or single nucleotide polymorphism/

5 factor v leiden mutation.ab,kw,ti.

6 "factor v leiden".ab,kw,ti.

7 FVL.ab,kw,ti.

8 Activated protein C Resistance.ab,kw,ti.

9 APCR.ab,kw,ti.

10 R506Q.ab,kw,ti.

11 G1691A.ab,kw,ti.

12 single nucleotide polymorphism?.ab,kw,ti.

13 SNP.ab,kw,ti.

14 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13

15 thromboembolism/ or venous thromboembolism/ or thrombosis/ or venous thrombosis/ or vein thrombosis/ or deep vein thrombosis/

16 Pulmonary Embolism/ or lung embolism/

17 "venous thrombo*".ab,kw,ti.

18 "Deep vein thrombo*".ab,kw,ti.

19 pulmonary embolism.ab,kw,ti.

20 "pulmonary embolis*".ab,kw,ti.

21 VTE.ab,kw,ti.

22 DVT.ab,kw,ti.

23 PE.ab,kw,ti.

24 "lung embolis*".ab,kw,ti.

25 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24

26 cohort studies/ or follow-up studies/ or longitudinal studies/ or prospective studies/ or longitudinal study/ or cohort analysis/ or follow up/ or prospective study/

27 cohort stud*.tw.

28 cohort analysis.ab,kw,ti.

29 "major clinical stud*".ab,kw,ti.

30 controlled study.ab,kw,ti.

31 26 or 27 or 28 or 29 or 30

32 14 and 25 and 31

33 remove duplicates from 32

1.2 List 2: Questions and levels of the adapted NOS scale.

Selection

D1) Representativeness of patients with FVL in the community

a) truly representative: Recruitment without knowledge of FVL. (+)

b) Selection according to the presence of FVL (-)

c) no description of the derivation of the cohort (?)

D2) Selection of the non-exposed cohort

a) drawn from the same study population as the exposed cohort (+)

b) drawn from a different source (-)

c) no description of the derivation of the non-exposed cohort (?)

D3) Ascertainment of exposure: Testing of FVL

a) APCR or/and PCR tested at baseline (+)

b) Patient documentation (+)

c) no description (?)

D4) Demonstration that recurrent VTE was not present at start of study

a) Objectively confirmed, described how recurrent VTE was handled to differentiate from initial VTE (+)

b) Objectively confirmed, unclear how recurrent VTE were handled to differentiate from initial VTE (?)

c) Not described, Questionnaire (-)

Comparability

D5) Comparability of cohorts on the basis of the design or analysis

a) Analyses were adjusted for confounders (+)

b) Analyses were not adjusted for confounders (-)

c) Not described (?)

Outcome

D6) Assessment of outcome: objectively confirmed VTE

a) independent blind assessment (+)

b) record linkage (+)

c) self-report (-)

d) no description (?)

D7) Acceptable length of follow-up: 2 years

a) yes (+)

b) no (-)

c) Unclear (?)

D8) Complete follow up

a) complete follow up (+)

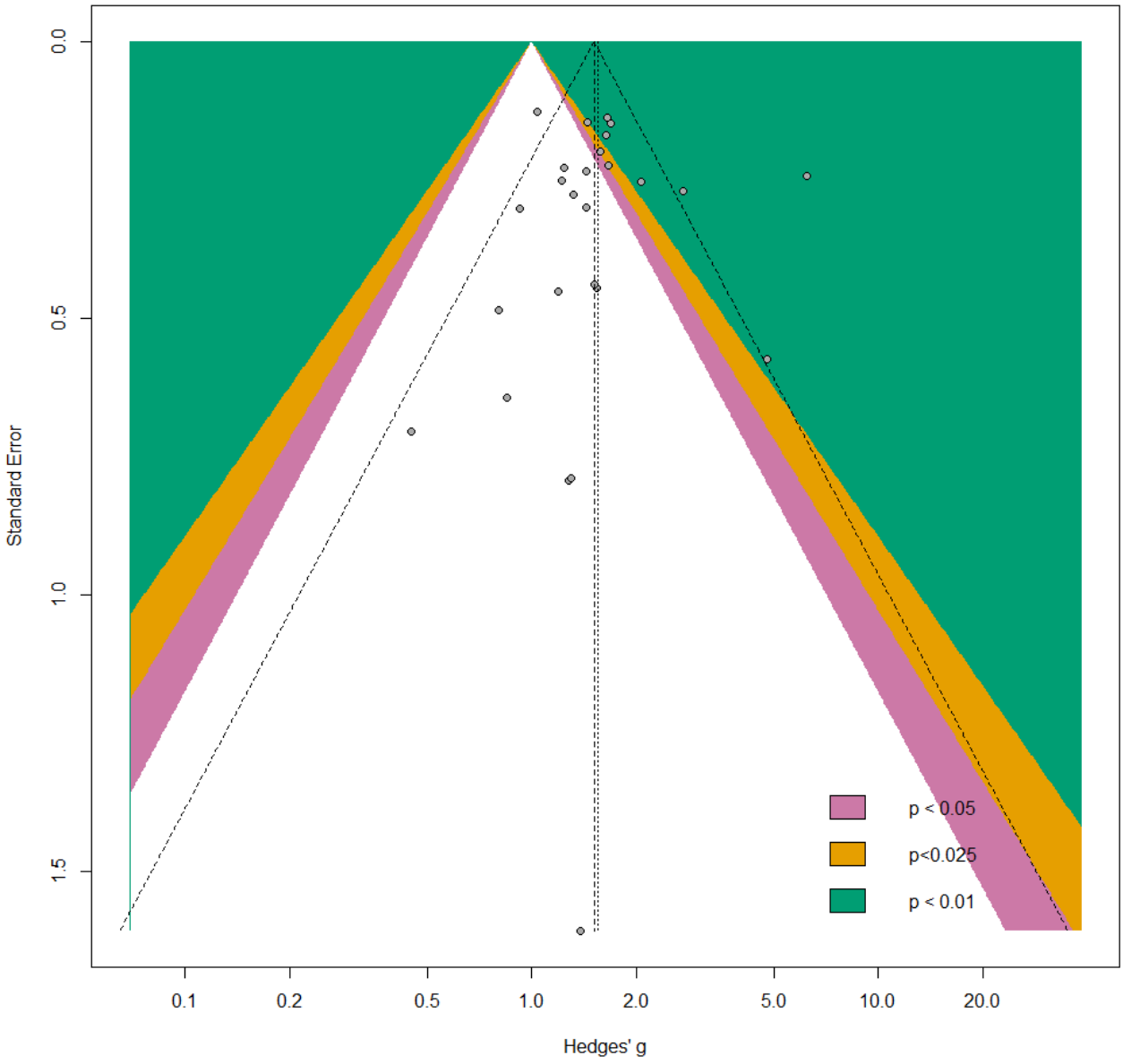
b) Small number lost < 10 % (+)

c) follow up rate < 90% (-)

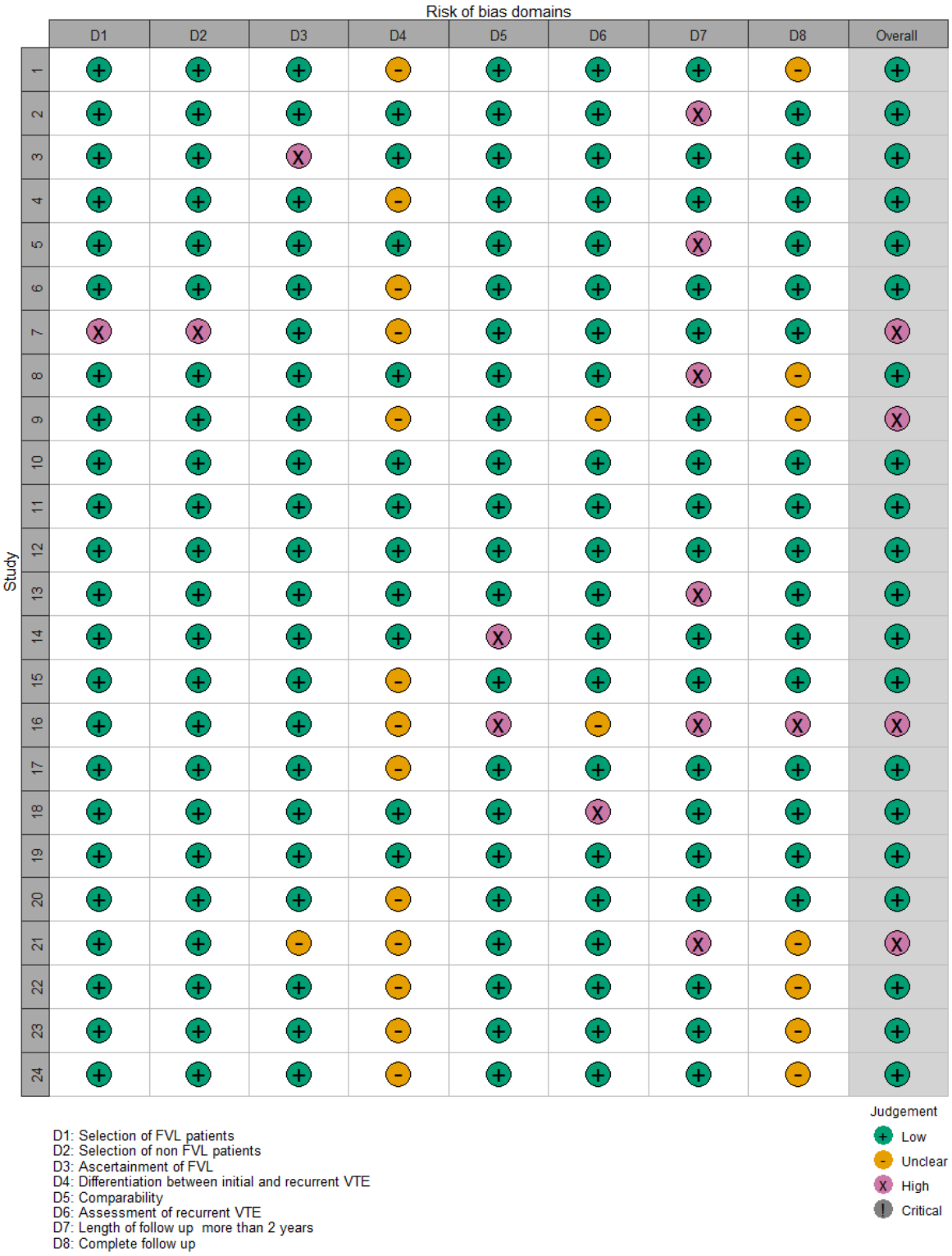
d) no statement (?)

2 Supplementary Figures and Tables

2.1 Figure S1: Funnel plot. Egger' test does not indicate the presence of a relevant asymmetry.



2.2 Table S1: Methodological quality of individual studies.



1. Simioni 1997(1)
2. Kearon 1999(2)

3. Lindmarker 1999(3)
4. Miles 2001(4)
5. Palareti 2003(5)
6. Christinasen 2005(6)
7. Vossen 2005(7)
8. Wahlander 2006(8)
9. Gonzalez-Porras 2006(9)
10. Prandoni 2007(10)
11. Poli 2007(11)
12. Eichinger 2008(12)
13. Rodger 2008(13)
14. Kearon 2008(14)
15. Chaireti 2009(15)
16. Obeidat 2010(16)
17. Sveinsdottir 2012(17)
18. Olie 2012(18)
19. Weingarz 2015(19)
20. Moreno 2016(20)
21. Bruzelius 2016(21)
22. Mean 2017(22)
23. Limperger 2020(23)
24. Hodeib 2021(24)

2.3 Table S2: Leave one out analysis

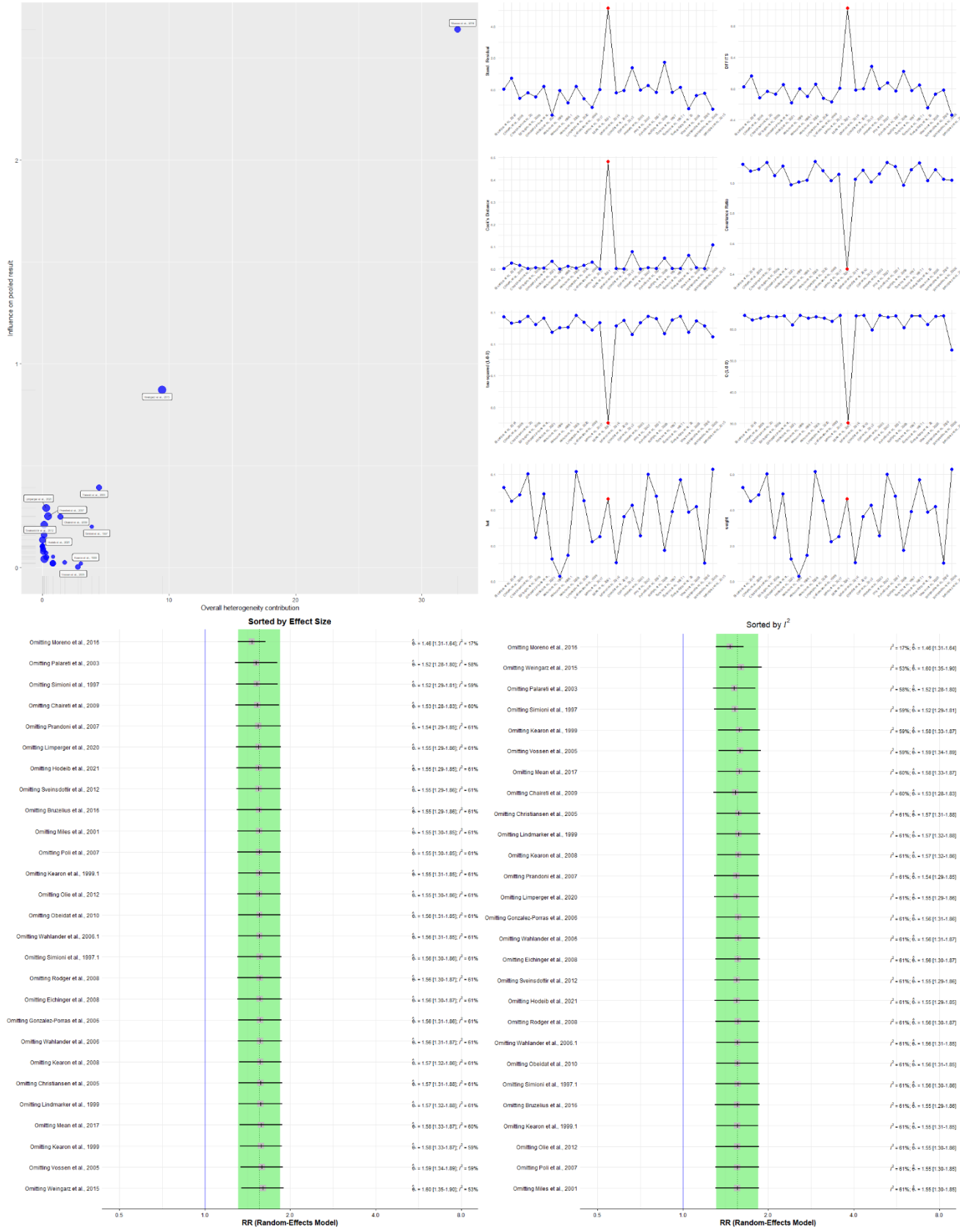
Study omitted	DFFITS	Cook's distance	Hat	Influence
1 - Simioni, 1997	0.220	0.048	0.018	
2 - Simioni, 1997	-0.170	0.001	0.039	
3 - Kearon, 1999	-0.183	0.033	0.013	
4 - Kearon, 1999	-0.003	0.000	0.003	
5 - Lindmarker, 1999	-0.123	0.016	0.045	
6 - Miles, 2001	0.003	0.000	0.025	
7 - Palareti, 2003	0.284	0.078	0.043	
8 - Christiansen, 2005	-0.121	0.015	0.048	
9 - Vossen, 2005	-0.247	0.060	0.039	
10 - Wahlander, 2006	-0.072	0.005	0.042	
11 - Wahlander, 2006	-0.021	0.000	0.010	
12 - Gonzales-Porras, 2006	-0.073	0.005	0.025	
13 - Prandoni, 2007	0.072	0.006	0.060	
14 - Poli, 2007	-0.001	0.000	0.026	
15 - Eichinger, 2008	-0.037	0.001	0.060	
16 - Rodger, 2008	-0.032	0.001	0.048	
17 - Kearon, 2008	-0.103	0.011	0.015	
18 - Chaireti, 2009	0.159	0.011	0.015	
19 - Obeidat, 2010	-0.020	0.000	0.010	
20 - Sveinsdottir, 2012	0.046	0.002	0.057	
21 - Olie, 2012	-0.005	0.000	0.036	
22 - Weingarz, 2015	-0.338	0.107	0.063	
23 - Moreno, 2016	1.028	0.481	0.046	*
24 - Bruzelius, 2016	0.019	0.000	0.053	
25 - Mean, 2017	-0.174	0.030	0.022	
26 - Limperger, 2020	0.057	0.004	0.061	
27 - Hodeib, 2021	0.052	0.003	0.049	

Cut-offs:

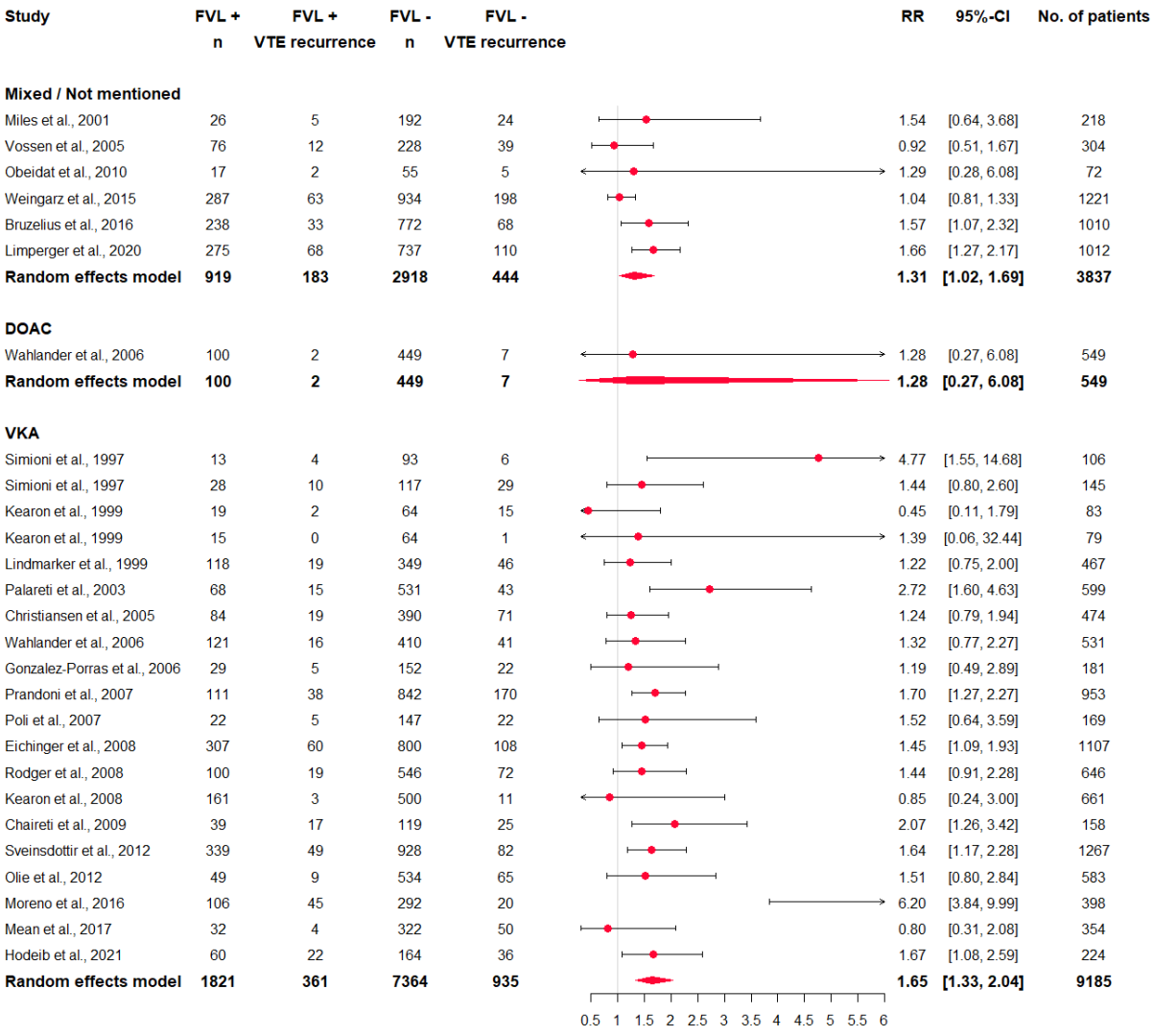
- The studentized residuals are outside -1 and 1
- The DFFITS value is larger than 3(pk-p)
- The area of the lower tail of a Chi-square distribution with p degrees of freedom cut off by the Cook's distance is larger than 50%

P = number of model coefficients; k = number of studies

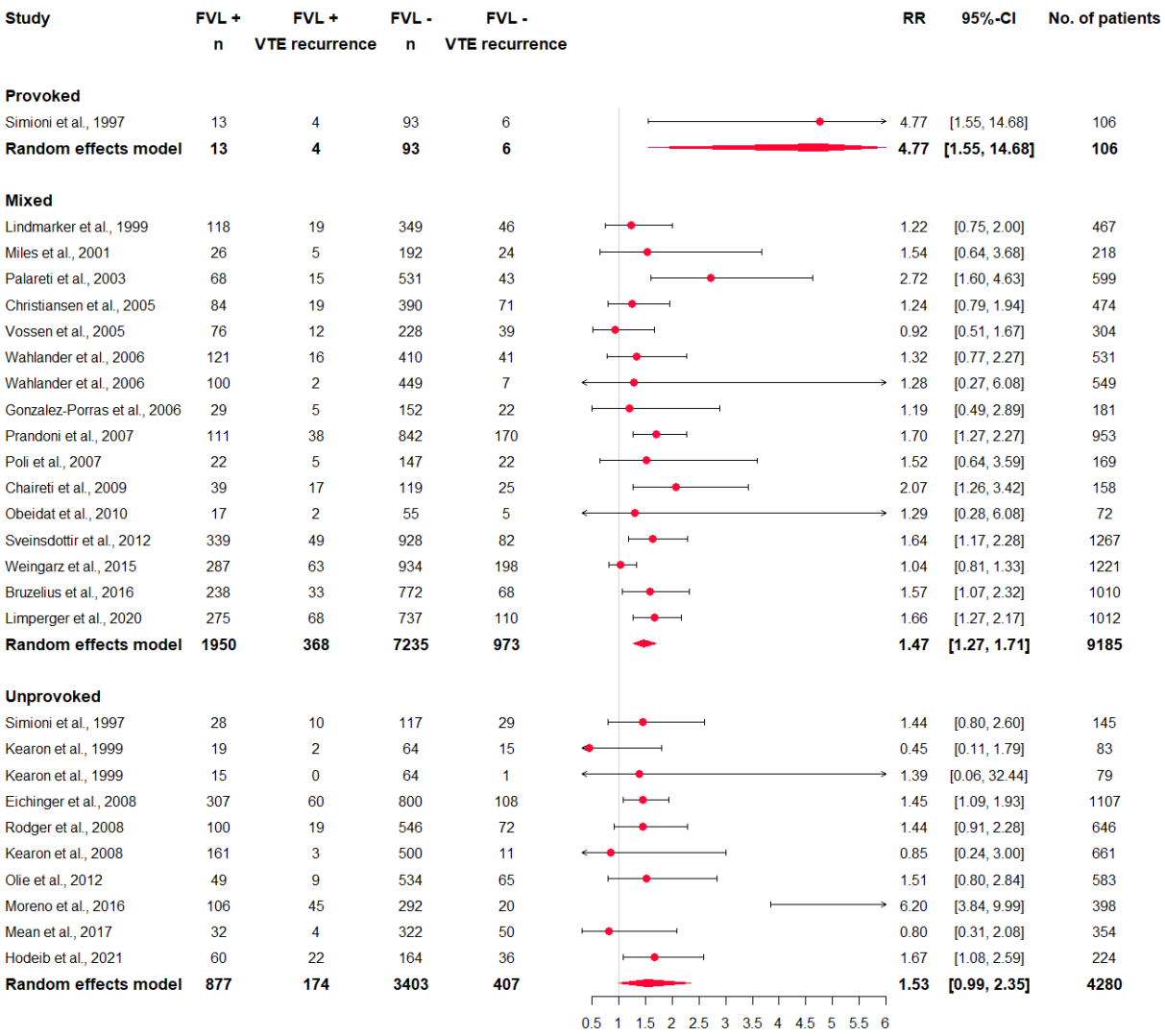
2.4 Figure S2: Influence analysis



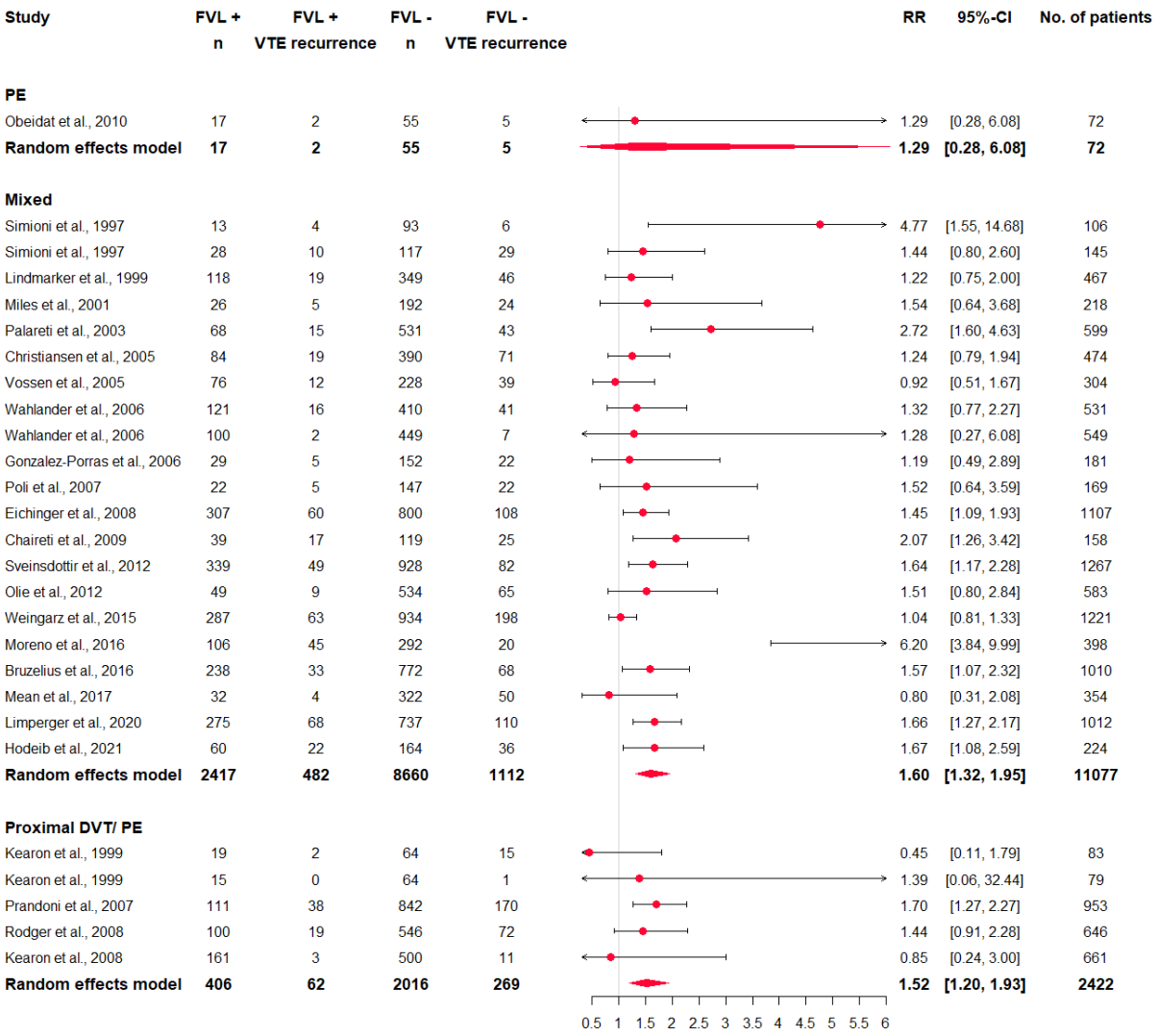
2.5 Figure S3: Subgroup analysis; type of anticoagulation



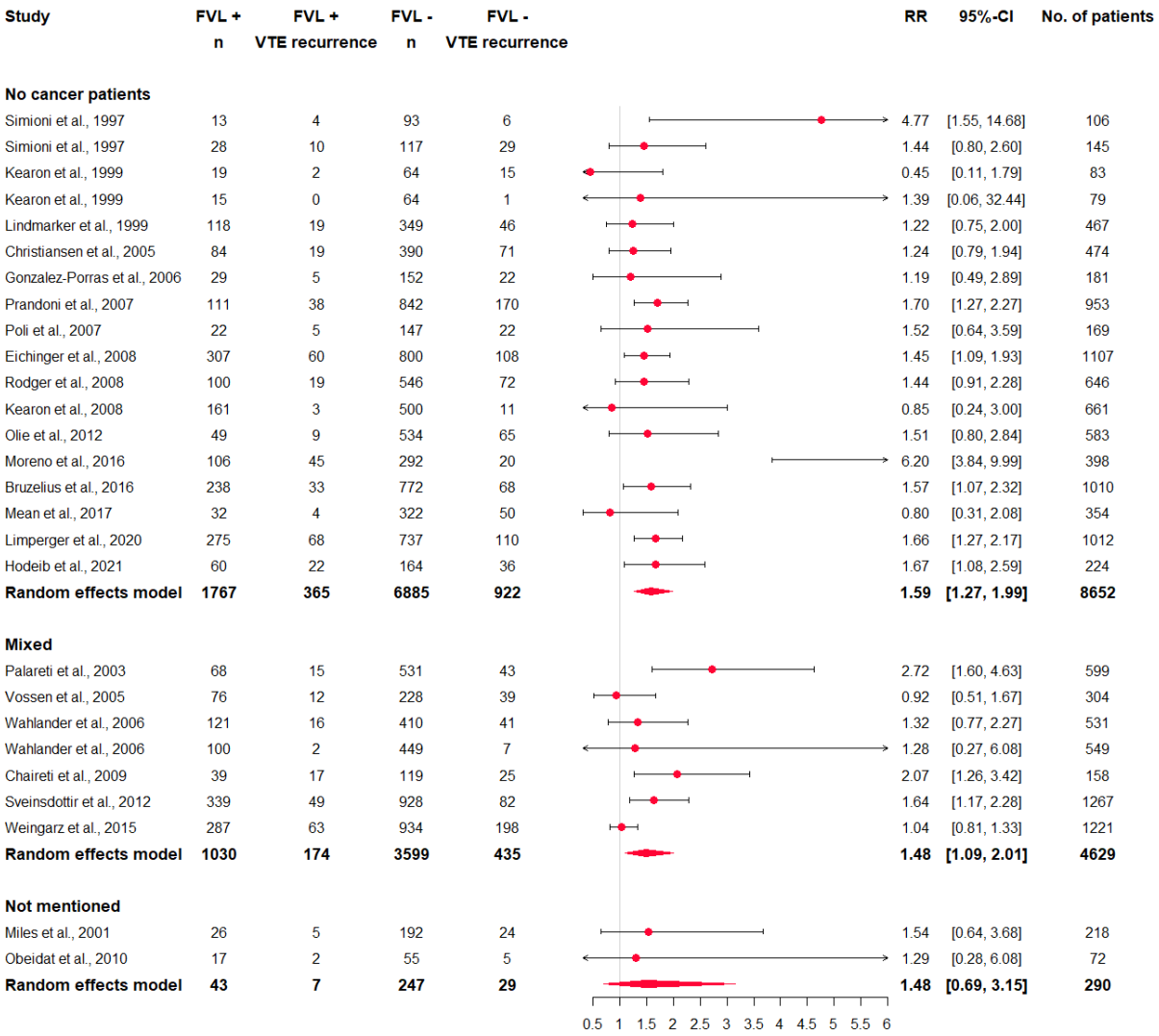
2.6 Figure S4: Subgroup analysis; trigger of initial event



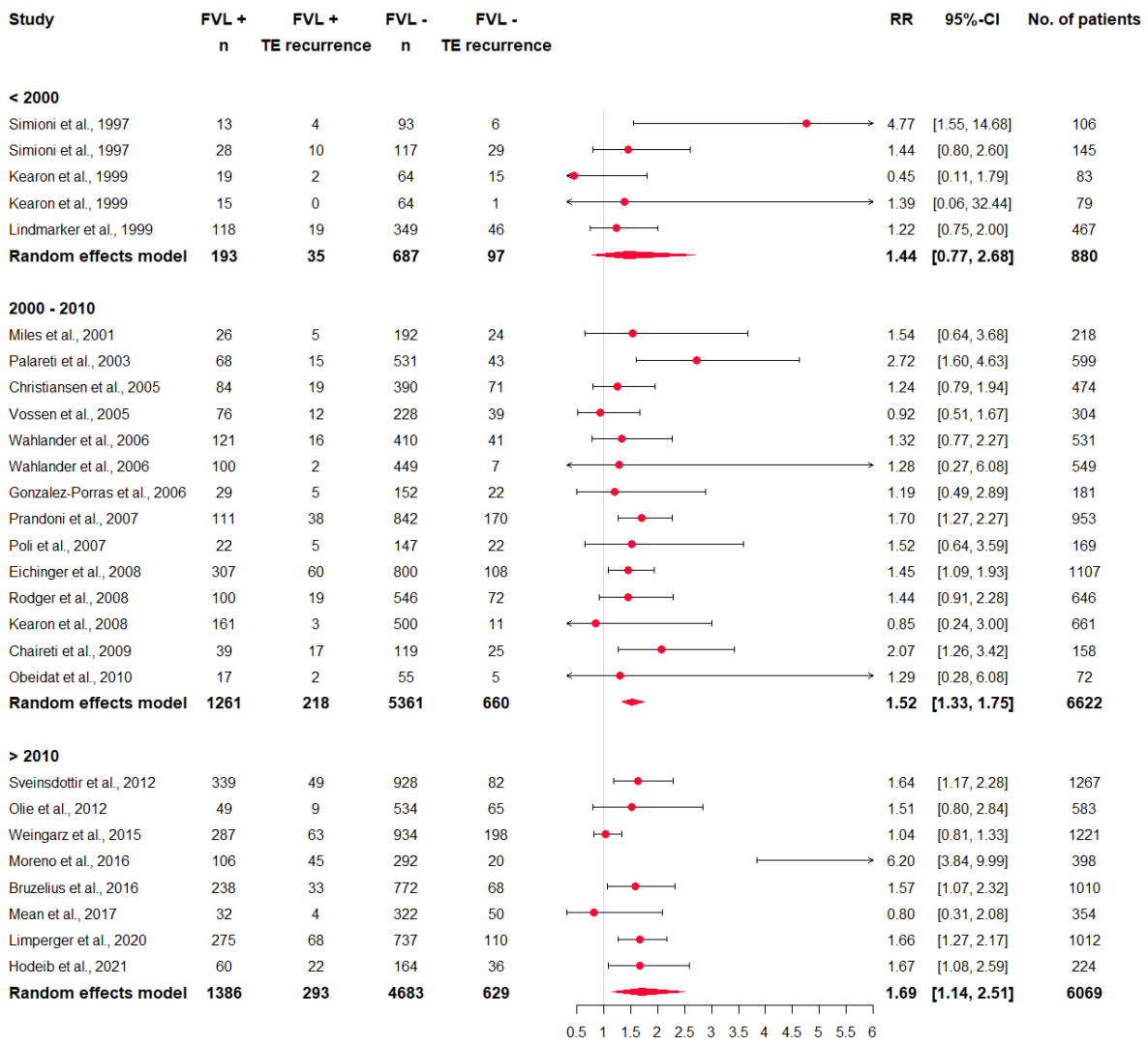
2.7 Figure S5: Subgroup analysis; localization of initial event



2.8 Figure S6: Subgroup analysis; with and without cancer patients



2.9 Figure S7: Subgroup analysis; year of publication

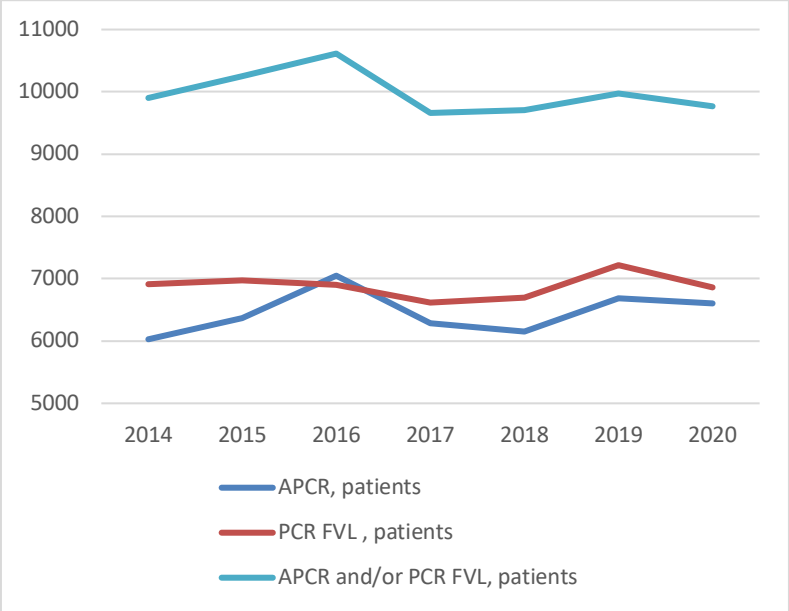


2.10 Table S3: Frequency of testing for FVL mutation in the Swiss health care system

		2014	2015	2016	2017	2018	2019	2020
		n (%)*	n (%)*	n (%)*	n (%)*	n (%)*	n (%)*	n (%)*
APCR (#1086.00)	patients	6028 (0.1)	6368 (0.1)	7049 (0.1)	6285 (0.1)	6156 (0.1)	6685 (0.1)	6601 (0.1)
	total	6206	6574	7206	6484	6298	6944	6810
PCR FVL mutation (#6200.64)	patients	6915 (0.1)	6976 (0.1)	6905 (0.1)	6617 (0.1)	6697 (0.1)	7218 (0.1)	6864 (0.1)
	total	7118	7187	6980	6793	6923	7614	7010
Total number of patients with APCR and/or PCR FVL		9900 (0.2)	10249 (0.2)	10614 (0.2)	9661 (0.2)	9706 (0.1)	9976 (0.2)	9768 (0.1)

* percentage of the Swiss population

2.11 Figure S8: Variation in FVL mutation testing from 2014 to 2020



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